Bulkstream

Developer Test: Edible Oil Tonnage Calculator

Overview

You are required to build a simple web application that calculates the tonnage (MT) of edible oil based on user-submitted volume, density, and temperature, using a Volume Correction Factor (VCF) retrieved from a table named 'vcftable'

Objective

Create a form that:

- Accepts volume(litres), density(kg/m3) and temperature (C)
- Retrieves the Volume Correction Factor (VCF) from the vcftable
- Calculates the tonnage using the formula;
  Tonnage(MT) =(Volume * Density * VCF)/1000
- Stores the result in a database table
- Displays the computed tonnage and a summary

Database setup

1. Import the provided SQL file vcftable.sql into your database. This table contains the volume correction factors indexed by density to the nearest 0.5 kg/m3 and temperature to the nearest 0.25C
2. Create a table named oil_tonnages to store the results

Output

- Display the calculated tonnage and VCF used
- Display a list of historical calculations with sort and search

Bonus Points

- Form validation with descriptive errors
- Tailwind/Bootstrap styling

Deliverables

- Github repository link
- README.md with setup instructions
- (Optional): Deployed demo link